Measurement Changes Phenomenon Student Handout

TOK Cognitive Science Activity Bank • Natural Sciences • 25-35 min

Category	Natural Sciences	**Time**	25-35 min
Difficulty	Easy	**ID**	measurement-changes-phenomenon

Big idea: Measurement does not merely capture phenomena; it operationalizes them.

Student-Facing Prompt

Inspect Measurement Changes Phenomenon Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Measurement Changes Phenomenon Stimulus A	Student-facing stimulus 1: Measure classroom attention by eyes on speaker, notes taken, response accuracy, or self-reported focus. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Measurement Changes Phenomenon Stimulus B	Student-facing stimulus 2: Measure noise by peak loudness, average loudness, teacher annoyance, or number of interruptions. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Measurement Changes Phenomenon Reveal / Twist	Student-facing stimulus 3: Measure noise by peak loudness, average loudness, teacher annoyance, or number of interruptions. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Measurement does not merely capture phenomena; it operationalizes them.

Actual Lesson Questions

#	Question for students
1	After inspecting Measurement Changes Phenomenon Stimulus A, what is your first knowledge claim?
2	Which exact detail from Measurement Changes Phenomenon Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Does measurement reveal or define what is being studied?

Ready-to-Use Examples

• Measure classroom attention by eyes on speaker, notes taken, response accuracy, or self-reported focus.

• Measure noise by peak loudness, average loudness, teacher annoyance, or number of interruptions.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Activity Steps

1. Choose one phenomenon and ask groups how it could be measured.
2. Assign different tools or operational definitions to different groups.
3. Groups collect quick data using their assigned method.
4. Compare results and explain why the same phenomenon produced different data.
5. Debrief the relationship between measurement, concepts, and evidence.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

• Knowledge question: Does measurement reveal a phenomenon or partly define it?

• When do different measures count as evidence for the same thing?

• How do instruments extend and constrain scientific knowledge?

Knowledge Questions

• Does measurement reveal or define what is being studied?

• How do instruments shape scientific evidence?

• What makes one measure more valid than another?

Transfer Example

For example, run this activity with a behaviour that changes when students know it is being measured. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

• One thing I first treated as evidence was...

• My confidence changed because...

• A better knowledge question I can now ask is...